



SketchFlex:

Facilitating Spatial-Semantic Coherence in Text-to-Image Generation with Region-Based Sketches

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Text-to-Image



“A dog standing on the Time Square at night...”



“A painting typically depicts a vast, romanticized landscape...”

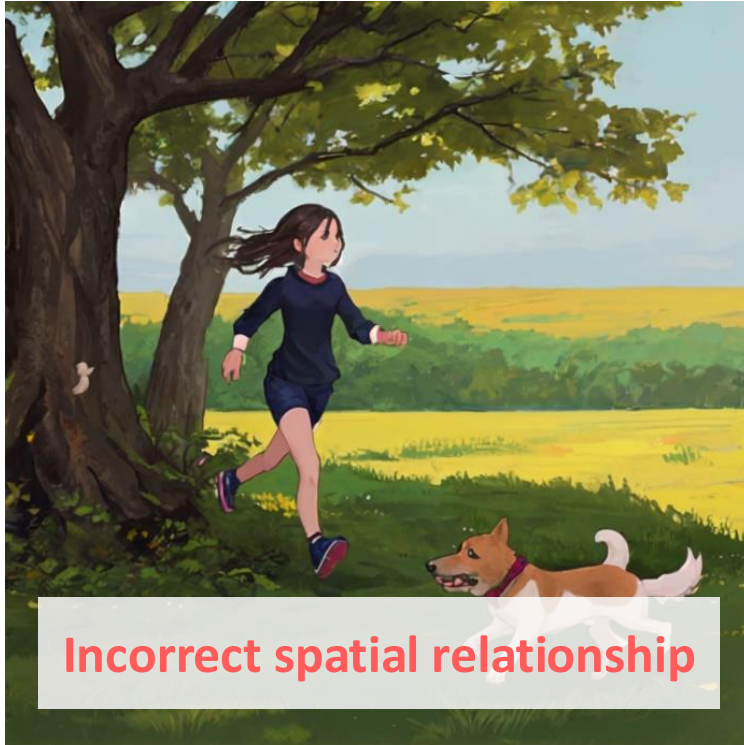
Background

Formative study

Method

Evaluation

Text-to-Image: Text control



Incorrect spatial relationship

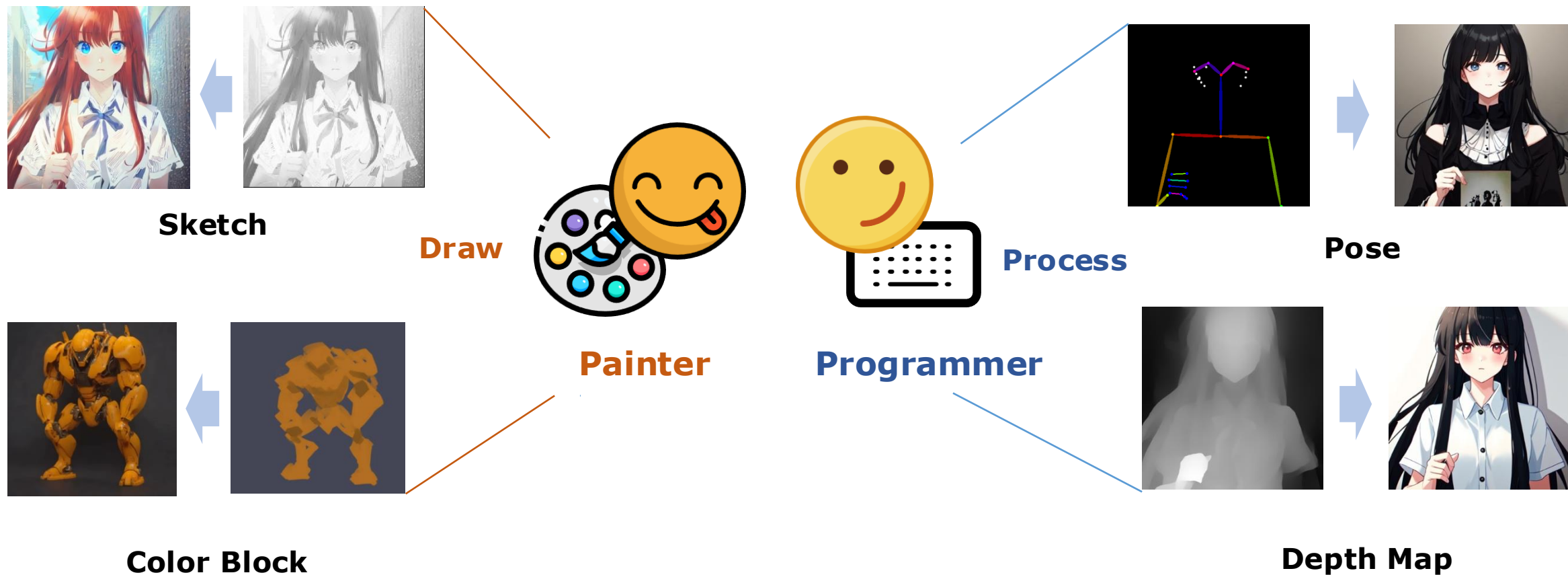
*"A girl **siting** on the **left** of a tree, a dog running on the **right** of the tree..."*



Incorrect object number

*"**Two child** is playing with **one dog**, an old man sitting near the lake..."*

Text-to-Image: Fine-grained control



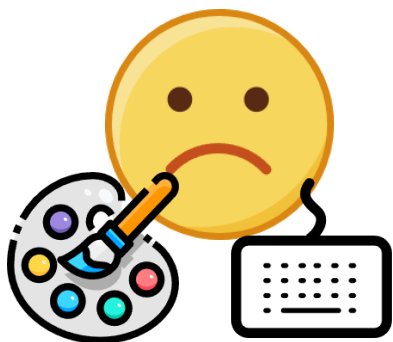
Background

Formative study

Method

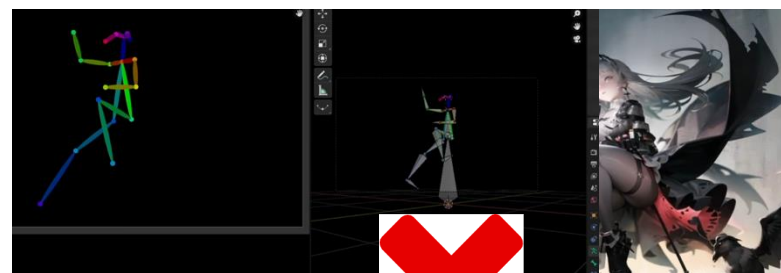
Evaluation

Text-to-Image: Fine-grained control

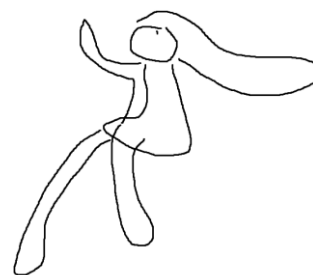


Novice User

*Requires
expertise
&
extensive
learning*



Professional Software



Rough Sketch



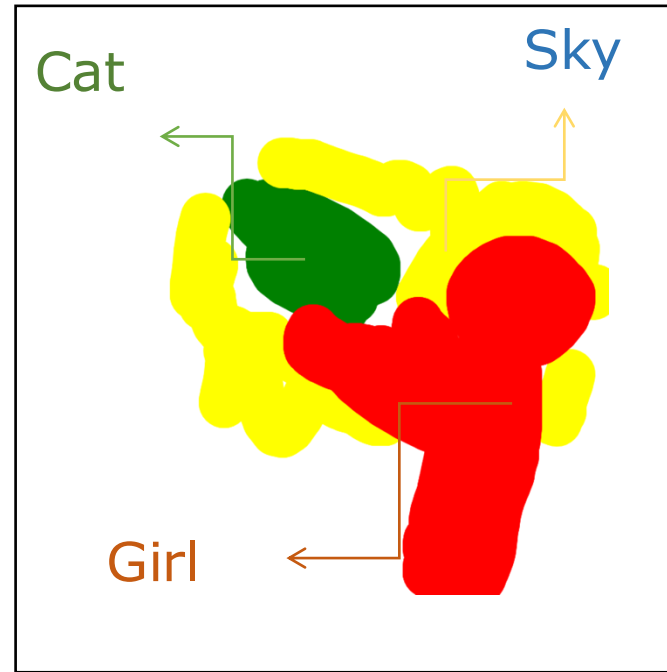
Undesired Result

Text-to-Image: Region-based control



Novice User

*Draw
rough
regions*



*Region-
based
generation*

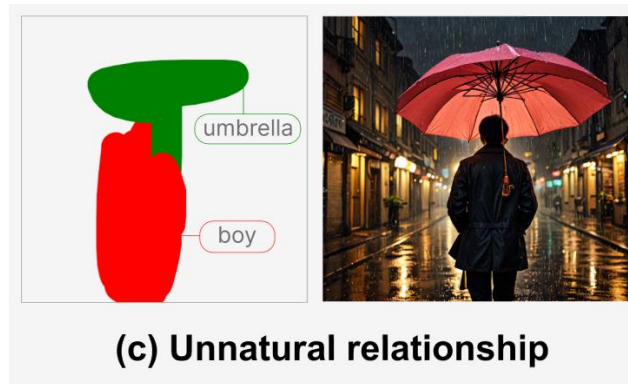


Region-based control challenges

1. Misalignment between sketch and prompt



2. Inappropriate prompt for multiple objects



Region-based control challenges

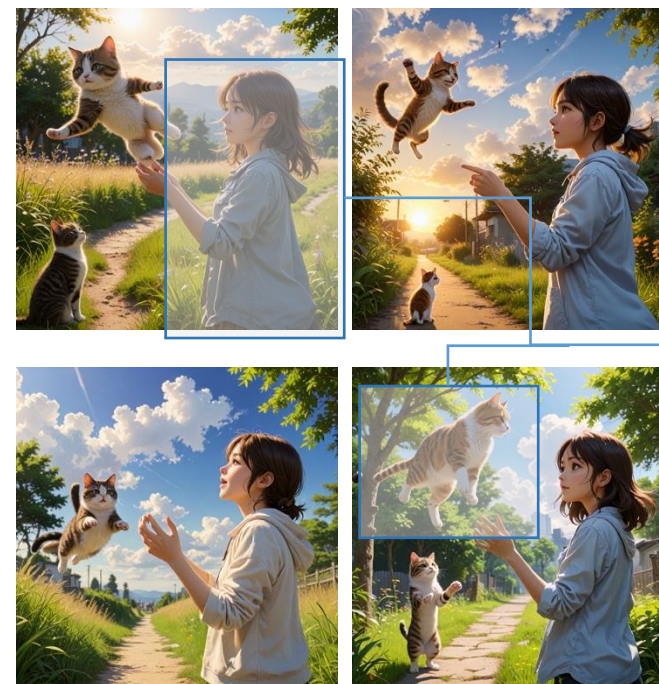


Novice User

Rough Sketch



Rough region cause large variety of each generated object



Preferred

Hard to fix the desired shape for each object in end-to-end generation

Formative study

Participants: Eight GenAI users (4F, 4M, aged 25–33), Four were novices (<6 months), and four were experts (>1 year). Painting skills ranged from no experience (3) to beginner(1) and proficient (4).

Procedure: 60–90 min interviews covered GenAI experience, live demos with Stable Diffusion/Midjourney/ComfyUI, and discussions on use cases, limitations, and future expectations.

Formative study - findings

- F1.** Rough sketch based spatial control is preferable by novice users.
- F2.** Combining prompt tuning and spatial control is challenging.
- F3.** Iterative generation and refinement is difficult in end-to-end generation.

Formative study - design goals

1. Rough sketch based spatial control is preferable by novice users.

G1. Providing Flexible Spatial Control of T2I Generation

2. Combining prompt tuning and spatial control is challenging.

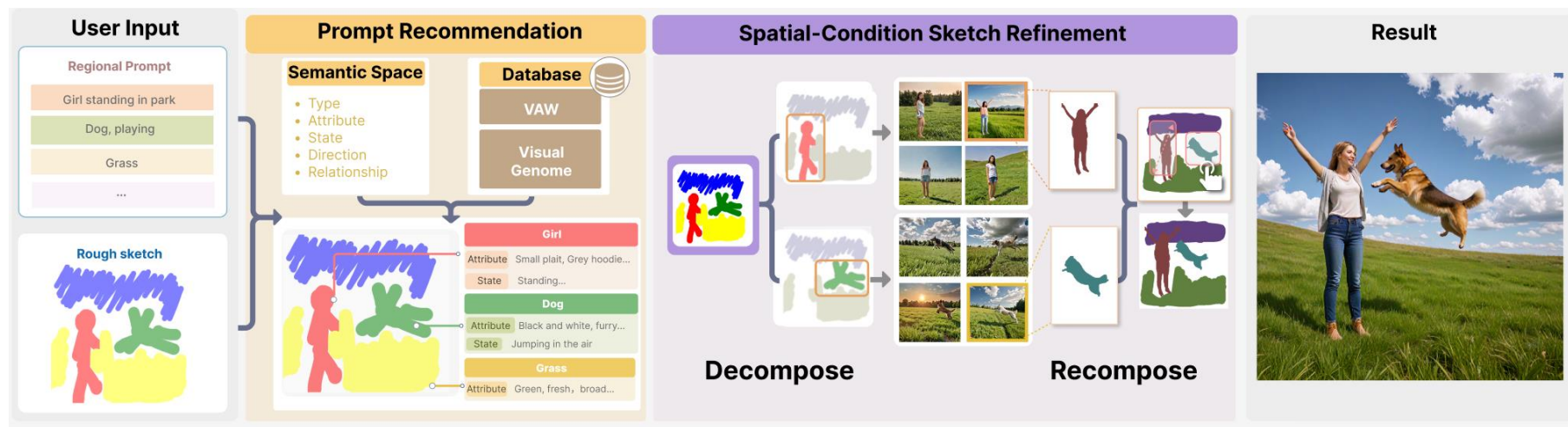
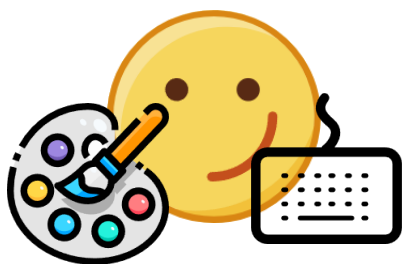
G2. Automating Prompt Tuning for Spatial Conditioned T2I Generation.

3. Iterative generation and refinement is difficult in end-to-end generation.

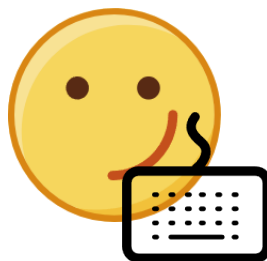
G3. Enhancing Flexibility in Image Adjustment by Decomposed Generation.

Method

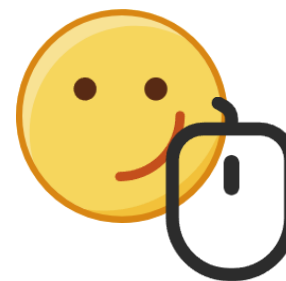
Draw Sketch &
input simple prompt



Adjust automatically
recommended Prompts



Select and adjust single
object shape



Background

Formative study

Method

Evaluation

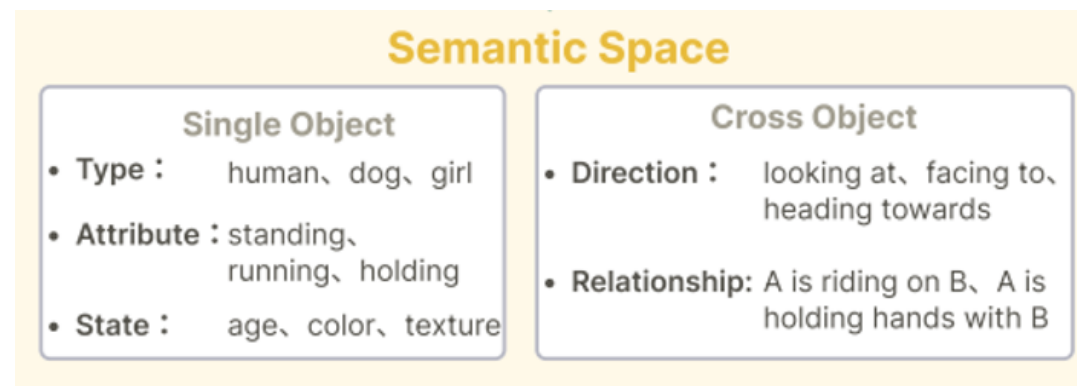
Method

Identify the types of prompts needed to generate a coherent image under rough sketch-based control.

- Analyzing online sources
- Prompt-guideline literature
- Image description datasets

Single Object:Type, Attribute, State,

Cross Object: Direction, Relationship



Method

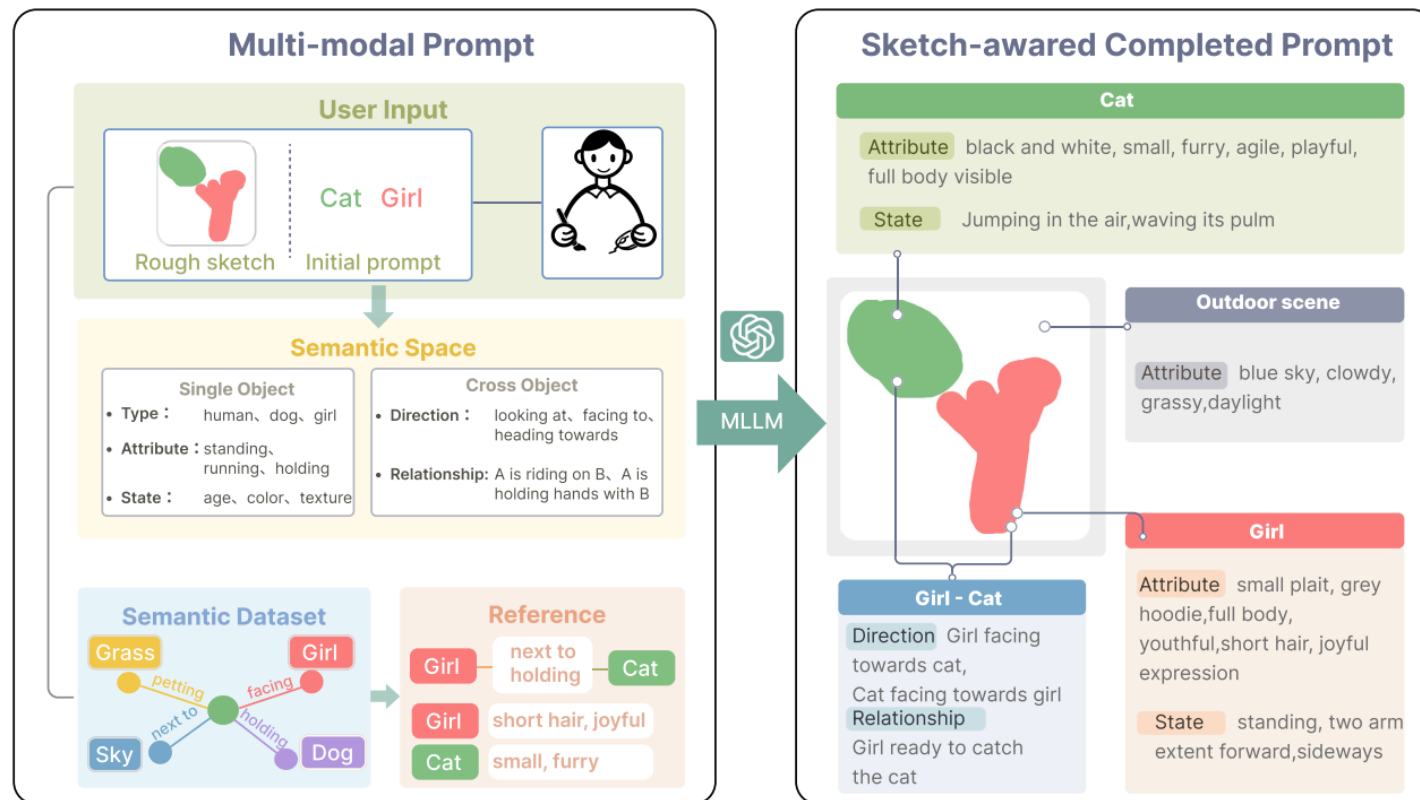
User input **rough sketch** and **prompt**



Retrieve **semantic reference** from realworld dataset



Complete region prompts within a **curated semantic space** using a **MLLM** (GPT4o)

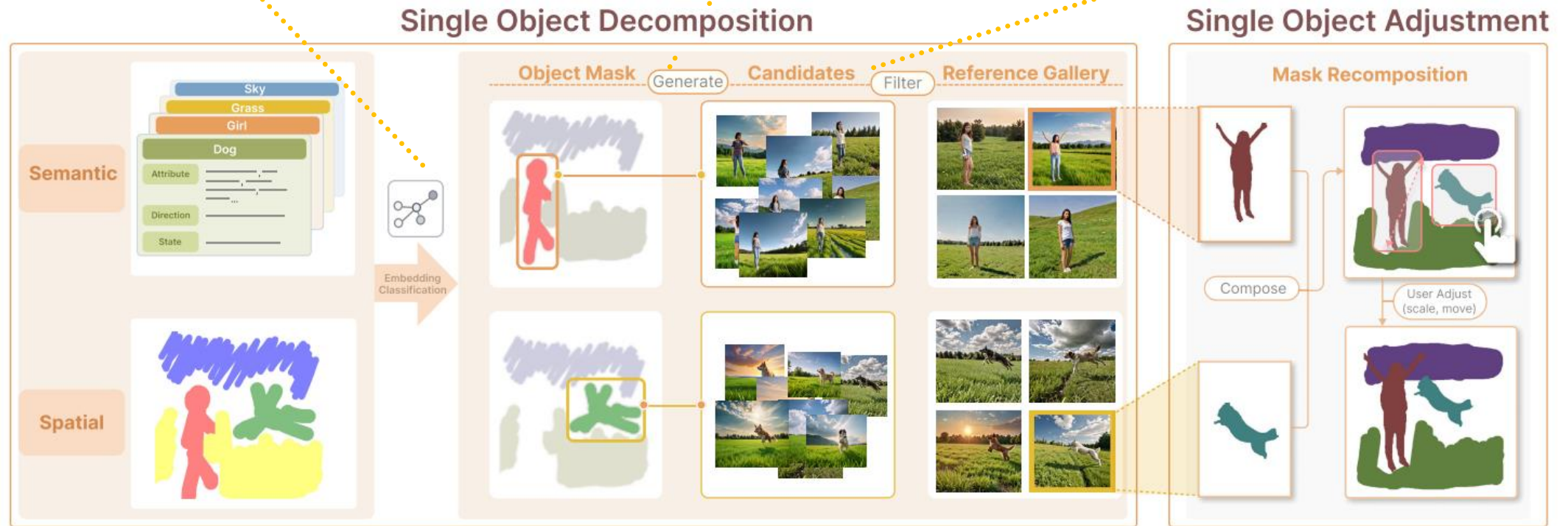


Method

Text embedding similarity based on COCO-stuff

Lightning model with low resolution

Weighted clip score + IoU score



Evaluation

Participants:

- GenAI experience: 9 novices, 3 experienced.
- Painting skills: 5 novices, 5 with basic skills, and 2 proficient.

Baselines:

- Text-to-Image
- Dense Diffusion

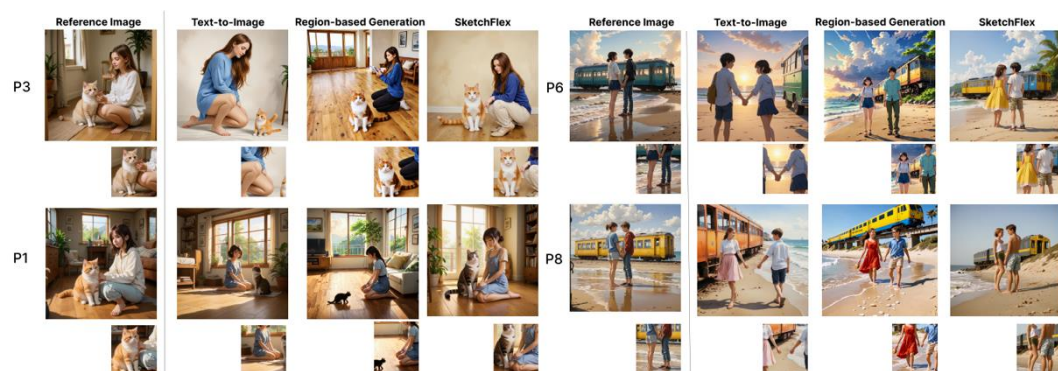
Tasks:

- Close-ended task
- Open-ended task

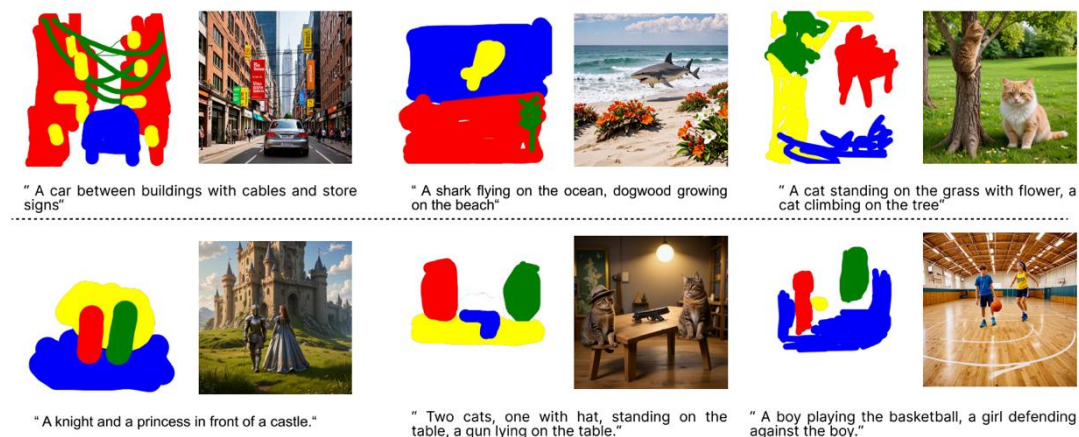
Evaluation

Quantitative: IoU score between object pairs, feature & feature ratings.

Qualitative: Feedback and discussions.



Close-ended tasks examples



Open-ended task examples

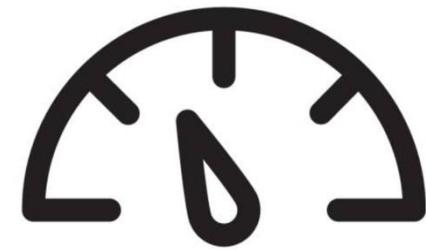
Discussion



Novice vs Expert



Applicability



Limitations

Background

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Method

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<https://github.com/SellLin/SketchFlex>